

## 3.2 - Selecting Samples

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## Overview

- Here we consider what to do when it is too difficult to generate all possible samples.
  - We examine the effect of using a randomly selected subset of all possible samples.
- 

Interest still lies in summarizing a population  $\mathcal{P}$  with an attribute  $a(\cdot)$  but now we do so by calculating the value of that attribute on a sample  $\mathcal{S} \subset \mathcal{P}$ :

$$a(\mathcal{S}) = \widehat{a(\mathcal{P})}$$

However, we must now acknowledge the existence of **sample error** and the possibility that:

$$a(\mathcal{S}) \neq a(\mathcal{P})$$

For any particular sample

- the attribute calculated based on the sample *could be* identical to the population attribute, or
- it might be so different that we would be completely misled about the true nature of the population attribute

This is why it is important to think carefully about **how** to select the sample and – if possible – do so in such a way to mitigate sample error.

- Even when the latter is possible, enormous care must be taken so that our own prejudices and pre-conceptions about the population do not render a sample that is misleading.

Given the reality that is sample error, it would be nice to understand the **magnitude** of error that can be expected.

- The **sampling distribution** of the attribute  $a(\mathcal{S})$  gives insight into this. Properties of this distribution can be determined
  - **exactly**, when all possible samples are available
  - **approximately**, when a subset of all possible samples is considered
  - **in expectation**, when a probabilistic sampling mechanism is used to draw a single sample
- We’ve already considered the first case. We briefly discuss the second below, but the majority of this section is devoted to third case – which is the most realistic of the three.

### 3.2.0 Randomly Selecting $m$ Samples

Consider drawing samples of size  $n$  from the population  $\mathcal{P}$ . The population  $\mathcal{P}_{\mathcal{S}}$  of *all such samples* has size  $M = \binom{N}{n}$  and is denoted

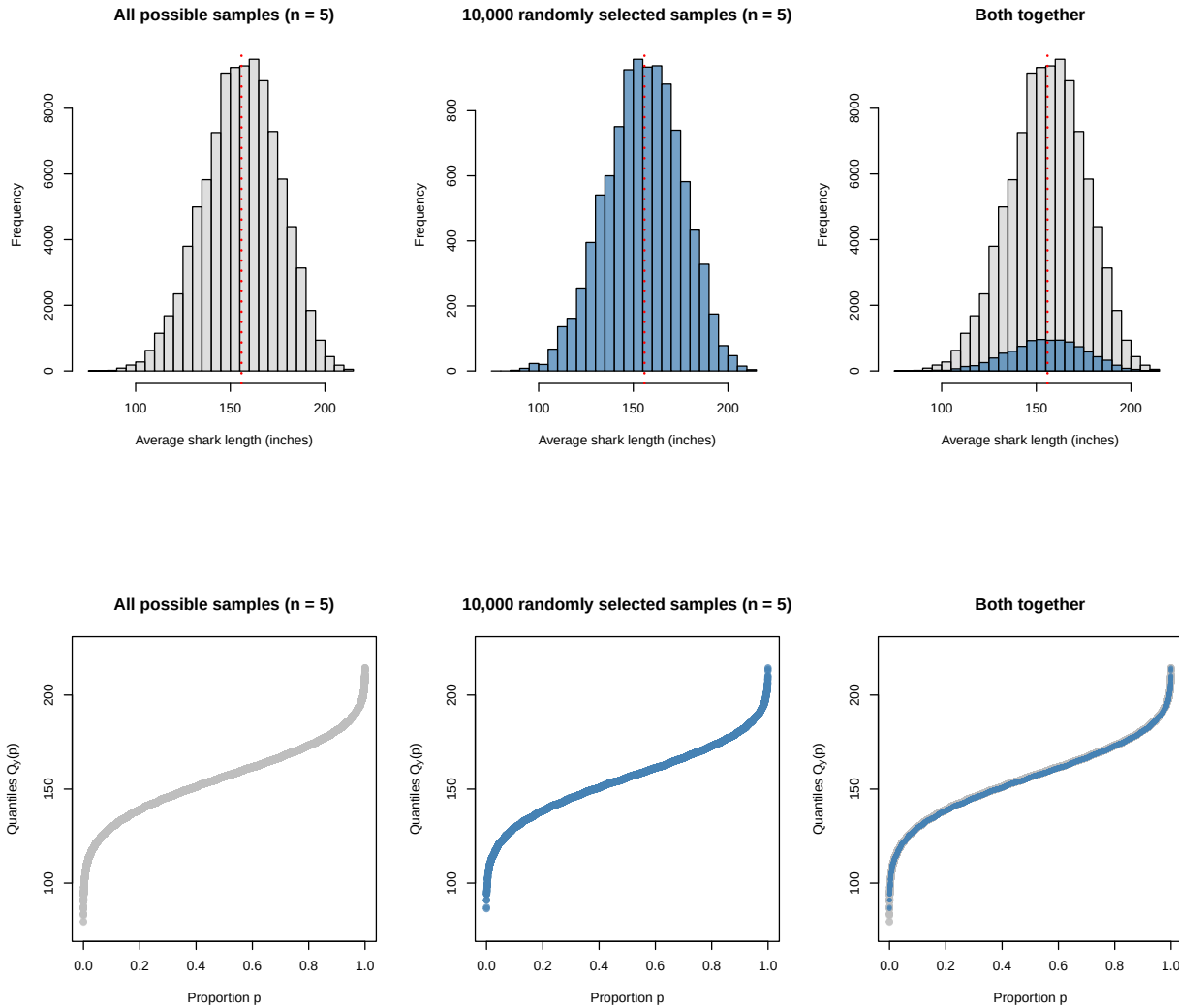
$$\mathcal{P}_{\mathcal{S}} = \{\mathcal{S}_1, \mathcal{S}_2, \dots, \mathcal{S}_M\}$$

- Any attribute  $a(\mathcal{S}_i)$  is now just a variate on that unit!
  - We then have a population of attributes

$$\mathcal{P}_{a(\mathcal{S})} = \{a(\mathcal{S}_1), a(\mathcal{S}_2), \dots, a(\mathcal{S}_M)\}$$

In general, it can be computationally expensive to calculate all possible samples.

- Instead, suppose we randomly select  $m$  samples  $\mathcal{S}_{u_1}, \dots, \mathcal{S}_{u_m}$  from the population  $\mathcal{P}_{\mathcal{S}}$  of  $M = \binom{N}{n}$  possible samples.
  - Let's revisit the Shark Data and in particular the subpopulation of shark encounters in Australian waters.
  - Here there are  $M = \binom{28}{5} = 98,280$  possible samples of size  $n = 5$
  - Let's consider a sample of  $m = 10,000$  of these



- What do we learn from these plots?

**Exercise:** Regenerate the plots above. Note that using the argument `add=TRUE` in the `hist` function will be handy.

- We can compare these graphs numerically using summary statistics.

```
temp = rbind(summary(avesSamp), summary(avesSampSel))
dimnames(temp)[[1]] = c("All Possible Samples", "10,000 Randomly Selected Samples")
temp
```

| ##                                  | Min. | 1st Qu. | Median | Mean     | 3rd Qu. | Max.  |
|-------------------------------------|------|---------|--------|----------|---------|-------|
| ## All Possible Samples             | 79.2 | 142.4   | 156.8  | 155.8929 | 169.8   | 214.4 |
| ## 10,000 Randomly Selected Samples | 86.4 | 142.0   | 156.2  | 155.4749 | 169.6   | 214.4 |

## Interesting Aside: The Distribution of a Histogram

- Suppose the histogram based on all possible samples has  $K$  bins

$$B_1 = (b_0, b_1], B_2 = (b_1, b_2], \dots, B_K = (b_{K-1}, b_K]$$

where the  $k^{\text{th}}$  bin  $B_k$  contains  $M_k \geq 0$  of the attribute values  $a(S_i)$   $i = 1, \dots, M$

- The bins contain the attribute values of all of the  $S_i \in \mathcal{P}_S$  so that  $\sum_{k=1}^K M_k = M$ .

- Now suppose that we select  $m$  samples at random from  $\mathcal{P}_S$  such that the probability that any given sample is selected is

$$p(S) = \frac{1}{M}$$

- Let  $m_k$  be the number of the  $m$  selected samples whose attribute value falls in  $B_k$ , with  $m = \sum_{k=1}^K m_k$ .

- With this notation,
  - the histogram based on all possible samples has heights  $M_1, \dots, M_K$  and
  - the histogram based on a sample of  $m$  of the possible samples has heights  $m_1, \dots, m_K$ .
- The probability of any particular histogram arising from a random selection of  $m$  samples is therefore a **multivariate hypergeometric** probability

$$\frac{\binom{M_1}{m_1} \binom{M_2}{m_2} \dots \binom{M_K}{m_K}}{\binom{M}{m}}$$

which, when  $m \ll M$  can be approximated by the **multinomial** probability

$$\binom{m}{m_1 \ m_2 \ \dots \ m_K} p_1^{m_1} p_2^{m_2} \dots p_K^{m_K}$$

with probabilities  $p_k = \frac{M_k}{M}$  for  $k = 1, \dots, K$ .

- From the multinomial, the expected value of the number of attribute values in each bin  $B_k$  is proportional to  $M_k$ . i.e.  $mp_k = \frac{m}{M}M_k$ .
  - The *frequency* histogram based on  $m$  samples is (in expectation) a scaled version of that of all possible samples.
  - The *density* histogram based on  $m$  samples is (in expectation) identical to that of all possible samples.

## Summary

- Randomly selecting a subset of all possible samples provides a reasonable approximation to the true sampling distribution of an attribute (determined by generating all possible samples).
- Therefore, we do not need to generate all possible samples to get an idea of the sample error or variability of the sample attribute values.

## Exercises

From the Samples exercises try problems 3 and 9.

### 3.2.1 Quantifying Sample Error

---

#### Overview

- We formalize the idea of the sampling distribution of an attribute.
  - To summarize the sampling distribution we use quantities such as sampling bias, sampling variance, and sampling mean squared error.
  - We illustrate these concepts by comparing different sampling designs on a toy example.
- 

In principle we select a sample  $\mathcal{S}$  from the population  $\mathcal{P}_{\mathcal{S}}$  containing all possible samples

- We do so with some probability  $p(\mathcal{S}) \geq 0$  of being selected. We require of course that

$$\sum_{\mathcal{S} \in \mathcal{P}_{\mathcal{S}}} p(\mathcal{S}) = 1.$$

- For any sample  $\mathcal{S} \in \mathcal{P}_{\mathcal{S}}$ , we have its **sample error**

$$\text{Sample Error} = a(\mathcal{S}) - a(\mathcal{P}).$$

- Recall, for any collection of samples (or population of samples)  $\mathcal{P}_{\mathcal{S}}$  containing  $M$  samples, we can calculate the **average sample error**

$$\text{Average Sample Error} = \frac{1}{M} \sum_{\mathcal{S} \in \mathcal{P}_{\mathcal{S}}} [a(\mathcal{S}) - a(\mathcal{P})].$$

We can quantify the concentration of sample errors in expectation using quantities such as **sampling bias**, **sampling variance**, and **sampling mean squared error**.

- By sampling  $\mathcal{S}$  randomly from  $\mathcal{P}_{\mathcal{S}}$  with probability  $p(\mathcal{S})$  we define the **sampling bias** as

$$\begin{aligned} \text{Sampling Bias} &= E[a(\mathcal{S})] - a(\mathcal{P}) \\ &= \sum_{\mathcal{S} \in \mathcal{P}_{\mathcal{S}}} a(\mathcal{S}) p(\mathcal{S}) - a(\mathcal{P}) \\ &= \sum_{\mathcal{S} \in \mathcal{P}_{\mathcal{S}}} [a(\mathcal{S}) - a(\mathcal{P})] p(\mathcal{S}) \end{aligned}$$

Sampling bias is just the **expected** sample error induced by the repeated random sampling of  $\mathcal{S}$  from  $\mathcal{P}_{\mathcal{S}}$ . If  $p(\mathcal{S}) = \frac{1}{M}$ , the sampling bias is identical to the average sample error of  $a(\mathcal{P})$ .

- The sampling bias depends on the attribute  $a(\cdot)$ , the set of possible samples  $\mathcal{P}_{\mathcal{S}}$ , and the sample probabilities  $p(\mathcal{S})$ .
- **Note:** If sampling bias is zero, then  $a(\mathcal{S})$  is called an **unbiased** estimator of  $a(\mathcal{P})$ .

- The **sampling variance** is defined as

$$Var[a(\mathcal{S})] = E \left[ (a(\mathcal{S}) - E[a(\mathcal{S})])^2 \right]$$

- This quantifies dispersion in the sample errors
- We may also use the **sampling standard deviation** defined as the square root of the variance

- Given a sample  $\mathcal{S}$ , we would like  $a(\mathcal{S})$  and  $a(\mathcal{P})$  to be as close as possible. We can use the **mean squared error** to quantify the expected squared distance between these two quantities

$$\begin{aligned} MSE[a(\mathcal{S})] &= E[(a(\mathcal{S}) - a(\mathcal{P}))^2] \\ &= Var[a(\mathcal{S})] + [\text{Sampling Bias}]^2 \end{aligned}$$

- Ideally, we would like to choose  $p(\mathcal{S})$  and/or  $\mathcal{P}_{\mathcal{S}}$ , so that both the square of sampling bias and the sampling variance are as small as possible.

**Note** that all expectations are taken with respect to the probabilities  $p(\mathcal{S})$  of choosing a sample  $\mathcal{S}$  from  $\mathcal{P}_{\mathcal{S}}$ .

## Attribute as an Estimator

- Thinking of the sampling distribution of an attribute  $a(\mathcal{S})$  gives rise to the notion of an attribute as an **estimator** (i.e., as a random variable).
- We can introduce a **random variable**, say  $A$ , that takes values  $a$  from the distinct values of  $a(\mathcal{S})$  for all  $\mathcal{S} \in \mathcal{P}_{\mathcal{S}}$ . The induced probability distribution is

$$\Pr(A = a) = \sum_{\mathcal{S} \in \mathcal{P}_{\mathcal{S}}} p(\mathcal{S}) \times \mathbb{I}(a(\mathcal{S}) = a)$$

- It follows that  $A$  is a discrete random variable.
- Probability statements about its values can be made using its distribution, including its expectation, variance, etc.
- Each of the definitions above (sampling bias, sampling variance, sampling MSE) can be defined in terms of this random variable and the corresponding probability distribution.

**Exercise:** Express the sampling bias, the sampling variance, and the sampling MSE in terms of this random variable.

## Example – Comparing Different Sampling Designs

In this toy example we illustrate two different **sampling designs** (i.e., two different ways of defining  $p(\mathcal{S})$  for  $\mathcal{S} \in \mathcal{P}_S$ ). We compare these methods of selecting  $\mathcal{S}$  on the basis of bias, variance and mean squared error.

- Suppose that the population  $\mathcal{P}$  consists of  $N = 5$  units:

```
set.seed(341)
pop5 = round(rnorm(5), 2)
pop5 = sort(pop5)
pop5 # our population
```

```
## [1] -1.06 -0.99 -0.31  0.83  0.87
```

- Now suppose we're interested in taking a sample of size  $n = 2$ . All possible samples of size 2 are:

```
sam2 = combn(5, 2)
colnames(sam2) = paste("S", 1:10, sep = "")
sam2
```

```
##      S1 S2 S3 S4 S5 S6 S7 S8 S9 S10
## [1,]  1  1  1  1  2  2  2  3  3  4
## [2,]  2  3  4  5  3  4  5  4  5  5
```

Note that the elements of the matrix above are the indices of the samples, not the actual values of the units.

- Next, we calculate the attribute (the mean) on these samples.

```
sam.avg <- apply(sam2, MARGIN = 2, FUN = function(s) {
  mean(pop5[s])
})
round(sam.avg, 3)
```

```
##      S1      S2      S3      S4      S5      S6      S7      S8      S9      S10
## -1.025 -0.685 -0.115 -0.095 -0.650 -0.080 -0.060  0.260  0.280  0.850
```

- Now consider two sampling designs:
  - d1: each sample is selected from the 10 possible samples with the same probability:  $p(\mathcal{S}) = \frac{1}{10} \quad \forall \mathcal{S}$ .
  - d2: different samples have different probabilities of being selected. There is no intuition behind this design, we use it simply for illustration.

```
d1 = rep(1/10, 10)
d2 = 2 * (abs(apply(sam2, 2, diff)) - 1)
d2 = d2/sum(d2)
designs = rbind(d1, d2)
colnames(designs) = paste("S", 1:10, sep = "")
round(designs, 2)
```

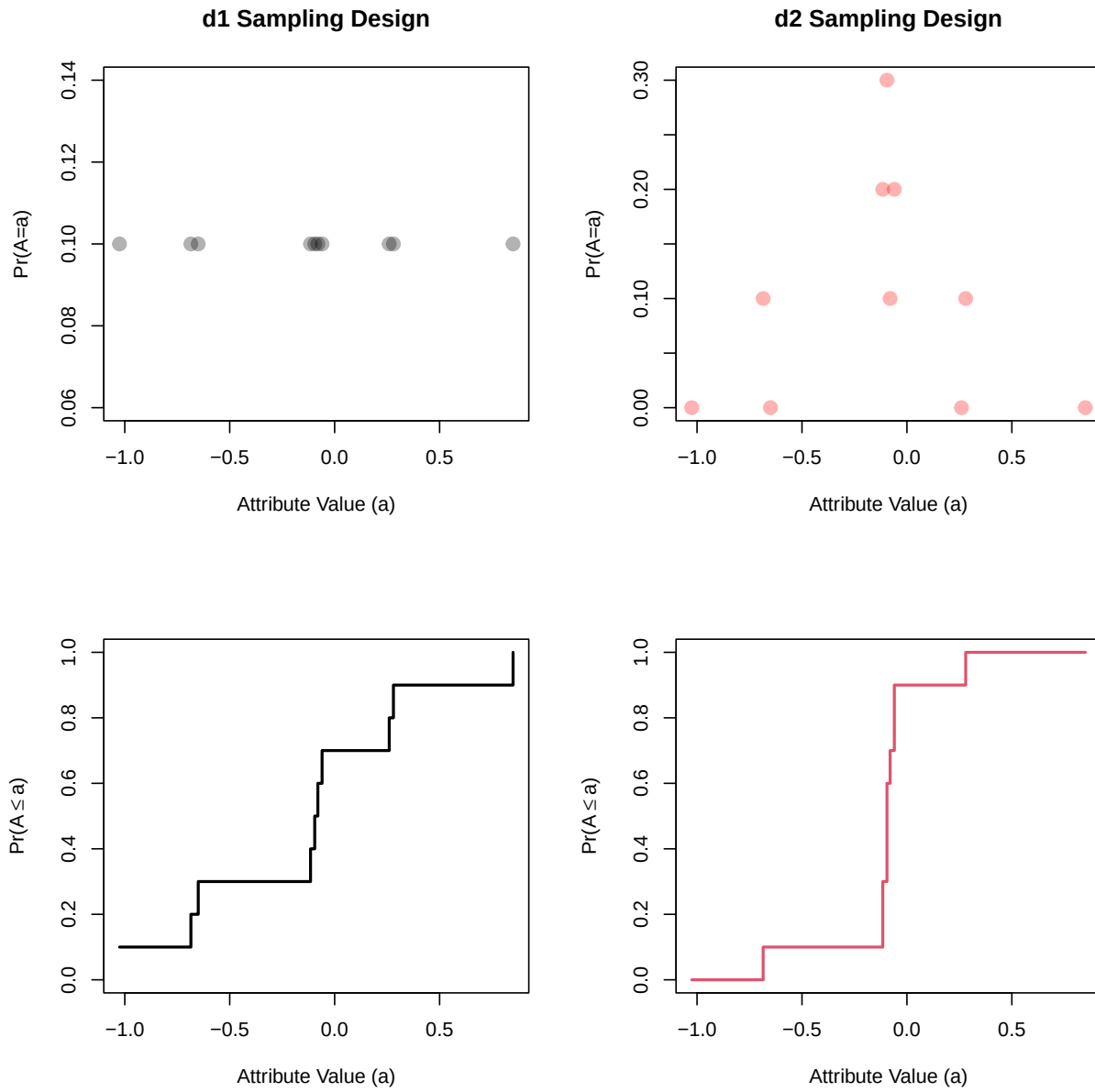
```
##      S1  S2  S3  S4  S5  S6  S7  S8  S9 S10
## d1 0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1
## d2 0.0 0.1 0.2 0.3 0.0 0.1 0.2 0.0 0.1 0.0
```

- The distribution of the attribute (here, the mean) induced by the sampling design, i.e.  $\Pr(A = a_i)$  is:

```
avg.ord = order(sam.avg)

par(mfrow = c(2, 2), oma = c(0, 0, 0, 0))
plot(sam.avg[avg.ord], d1[avg.ord], xlab = "Attribute Value (a)", ylab = "Pr(A=a)",
     pch = 19, main = "d1 Sampling Design", col = adjustcolor("black", 0.3),
     cex = 1.5)
plot(sam.avg[avg.ord], d2[avg.ord], xlab = "Attribute Value (a)", ylab = "Pr(A=a)",
     pch = 19, main = "d2 Sampling Design", col = adjustcolor("red", 0.3), cex = 1.5)

plot(sam.avg[avg.ord], cumsum(d1[avg.ord]), xlab = "Attribute Value (a)", ylab = expression("Pr(A" <=
     "a)"), pch = 19, type = "s", ylim = c(0, 1), lwd = 2)
plot(sam.avg[avg.ord], cumsum(d2[avg.ord]), xlab = "Attribute Value (a)", ylab = expression("Pr(A" <=
     "a)"), pch = 19, col = 2, type = "s", ylim = c(0, 1), lwd = 2)
```



- Note that the distribution of the attribute with respect to design d2 is more concentrated.

Let's evaluate the sampling error associated with the attribute (here, the mean) with respect to both sampling designs.

- We compare the two sampling designs d1 and d2 numerically using the sampling bias, sampling variance and sampling MSE

### Sampling bias

```
exp1 = sum(sam.avg * d1)
exp2 = sum(sam.avg * d2)

sam.bias = c(exp1, exp2) - mean(pop5)
round(sam.bias, 5)
```

```
## [1] 0.00 0.02
```

### Sampling Variance

```
sam.var = c(sum((sam.avg - exp1)^2 * d1), sum((sam.avg - exp2)^2 * d2))
round(sam.var, 5)
```

```
## [1] 0.26689 0.04893
```

### Sampling MSE

```
sam.MSE = sam.var + sam.bias^2
round(sam.MSE, 5)
```

```
## [1] 0.26689 0.04933
```

Alternatively, for our MSE calculation we could use the formula

$$MSE[a(\mathcal{S})] = E[(a(\mathcal{S}) - a(\mathcal{P}))^2]$$

```
MSE = c(sum((sam.avg - mean(pop5))^2 * d1), sum((sam.avg - mean(pop5))^2 * d2))
round(MSE, 5)
```

```
## [1] 0.26689 0.04933
```

**Question:** So which sampling design is better?

## 3.2.2 A Note on Large Populations

- As we've discussed, as the population size increases constructing all possible samples becomes prohibitive
  - For example, consider the agricultural census of US counties whose population consists of only  $N = 3078$  counties. For  $n = 100$ , there are  $\binom{3078}{100}$  or about  $1.4 \times 10^{190}$  possible samples.

- The combinatorial explosion is avoided if we examine only  $m$ , say  $m = 10,000$ , samples.
  - Unfortunately, if we have to enumerate all possible samples just to select from them we are no farther ahead.
- We have discussed methods of quantifying the distribution of sample errors, but at present these definitions assume that either  $\mathcal{P}_S$  is available or that  $\Pr(A = a)$  is known. Neither of which is very realistic.
- Considering all possible samples is a useful means to quantify sample error and a valid means for selecting a sample
  - But we require an approach to selecting samples and evaluating sample errors that reflects the realistic scenario in which *just one* sample is ever observed.

## Summary

- We formalized the idea of the sampling distribution of an attribute.
- The quantities sampling bias, sampling variance, and sampling mean squared error provide insight into the sampling distribution.
- They may be used to compare different sampling designs.

## Exercises

From the Samples exercises try problem 13. For continued learning try problems 5, 6 and 17.

### 3.2.3 Sampling Mechanisms

---

#### Overview

- We consider three different sampling mechanisms and compare them.
- 

Rather than select samples at random from all possible samples, the same outcome is effected by sampling the units that will appear in any particular sample

- In other words: rather than select  $\mathcal{S}$  with probability  $p(\mathcal{S})$  from  $\mathcal{P}_{\mathcal{S}} = \{\mathcal{S}_1, \mathcal{S}_2, \dots, \mathcal{S}_M\}$  we form  $\mathcal{S}$  by selecting  $n$  units  $u_{i_1}, u_{i_2}, \dots, u_{i_n}$  directly from the population of units  $\mathcal{P} = \{u_1, u_2, \dots, u_N\}$ .
- Each unit  $u$  in a sample  $\mathcal{S}$  is selected one at a time from the population  $\mathcal{P}$ .
- A *sequence* of the first  $k$  units  $u_i$  selected from  $\mathcal{P}$  is

$$s_k = (u_{i_1}, u_{i_2}, \dots, u_{i_k})$$

- A **sampling mechanism** is defined by the probabilities

$$\Pr(u) \quad \text{and} \quad \Pr(u \mid k, s_{k-1}).$$

where

- the first unit is selected with probability  $\Pr(u)$ , and
- the probability of the sequence of the first  $k$  units selected is

$$\Pr(s_k) = \Pr(u_{i_1}) \times \Pr(u_{i_2} \mid 2, s_1) \times \Pr(u_{i_3} \mid 3, s_2) \times \dots \times \Pr(u_{i_k} \mid k, s_{k-1}).$$

- To determine  $p(\mathcal{S})$  from a sampling mechanism:
  - Recognize that the order in which the units appear does not matter, i.e. any of the  $n!$  permutations of the elements of  $s_n$  counts as  $\mathcal{S}$
  - $p(\mathcal{S})$  is thus the sum of  $\Pr(s_n)$  over all permutations  $s_n$ .

## Simple Random Sampling Without Replacement (SRSWOR)

- The **sampling mechanism** is

$$\Pr(u) = \frac{1}{N} \quad \text{and} \quad \Pr(u \mid k, s_{k-1}) = \frac{1}{N - k + 1}$$

- The probability of the sequence  $s_n$  is

$$\Pr(s_n) = \frac{1}{N} \times \frac{1}{N-1} \times \frac{1}{N-2} \times \dots \times \frac{1}{N-n+1}$$

which is the same for all  $n!$  permutations, so

$$p(\mathcal{S}) = \frac{n!}{N(N-1)(N-2)\dots(N-n+1)} = \frac{1}{\binom{N}{n}}$$

This probability is the same as we had before for selecting  $n$  distinct units from a population of  $N$  distinct units.

- However, we now have a mechanism that allows us to select a sample *without first enumerating* all  $M = \binom{N}{n}$  possible samples in  $\mathcal{P}_S$ .
- In R, the indices of a simple random sample without replacement of size  $n$  from indices  $1, \dots, N$  is returned from the function call `sample(N,n)`.

```
set.seed(341)
pop15 = round(rnorm(15), 2)
pop15
```

```
## [1] -1.06 -0.31  0.87 -0.99  0.83  0.47 -0.66 -0.05  1.46 -0.72  0.82  1.34
## [13]  1.78 -1.12 -0.76
```

```
set.seed(341)
sample5 = sample(15, 5)
sample5
```

```
## [1]  9  4 13 11  1
```

```
pop15[sample5]
```

```
## [1]  1.46 -0.99  1.78  0.82 -1.06
```

- If rather than indices, the units were identified by the (assumed unique) contents of a vector `Pop`, then `sample(Pop,n)` would return the vector of units in the sample.

```
set.seed(341)
sample(pop15, 5)
```

```
## [1]  1.46 -0.99  1.78  0.82 -1.06
```

## Simple Random Sampling With Replacement (SRSWR)

- The **sampling mechanism** is

$$\Pr(u) = \frac{1}{N} = \Pr(u \mid k, s_{k-1})$$

and thus a sample  $S$  can contain one or more replicated units

- The probability of the sequence  $s_n$  is

$$\Pr(s_n) = \left(\frac{1}{N}\right)^n$$

- Note that unlike in the case of SRSWOR, we typically treat each  $s_n$  as an ordered sample and so

$$p(\mathcal{S}) = \Pr(s_n) = \frac{1}{N^n}$$

- The population of all samples  $\mathcal{P}_{\mathcal{S}}$  in this case contains  $M = N^n$  ordered samples.

- If we treat each  $s_n$  as an unordered sample similar to SRSWOR we obtain

$$p(\mathcal{S}) = \frac{\frac{n!}{n_1!n_2!\dots n_N!}}{N^n}$$

where  $n_u$  is the number of duplicates of unit  $u$  in the sample. e.g. if  $n_4 = 2$  then unit 4 was in the sequence (sample) twice.

- To generate simple random samples with replacement in R, we use `sample` as before except now with the argument `replace = TRUE` as in `sample(N, n, replace = TRUE)`.

```
set.seed(341)
pop15 = round(rnorm(15), 3)

set.seed(341)
sample5 = sample(15, 5, replace = TRUE)
sample5
```

```
## [1]  9  4 14 14 13
```

```
pop15[sample5]
```

```
## [1]  1.462 -0.993 -1.115 -1.115  1.783
```

```
set.seed(341)
sample(pop15, 5, replace = TRUE)
```

```
## [1]  1.462 -0.993 -1.115 -1.115  1.783
```

## A Weird Hybrid Sampling Mechanism (Let's call it SRSWH)

- The following mechanism was first explored by Basu (1958).
- Suppose we perform simple random sampling with replacement except that we **remove** any duplicate units.
  - The samples produced will have sizes anywhere from 1 to  $n$  according to how many distinct units were selected in a sample (sampling with replacement).

```
set.seed(341)
pop10 = round(rnorm(10), 3)
set.seed(341)
sample5 = sample(10, 5, replace = TRUE)
sample5
```

```
## [1]  9  4  1 10  1
```

- Simple random sample with replacement yields

```
pop10[sample5]
```

```
## [1]  1.462 -0.993 -1.060 -0.720 -1.060
```

- Simple random sample with replacement removing duplicate units yields

```
unique(pop10[sample5])
```

```
## [1]  1.462 -0.993 -1.060 -0.720
```

Note that since the number of duplicates is a random variable, the actual sample size ( $n$  minus the number of duplicates) is also a random variable here!

## Connections to Balls in an Urn

- Suppose that we have an urn containing  $N$  different balls that are either white or black.
  - We would like to estimate the proportion of balls in the urn which are black by drawing  $n$  balls at random from the urn.

### 1. Simple random sampling **without** replacement (SWSWOR).

- Randomly draw  $n$  balls from the urn one after another, **without replacing** any at any time.
- The estimate is the proportion of black balls in your sample.

### 2. Simple random sampling **with** replacement (SRSWR).

- Randomly draw  $n$  balls from the urn one after another, **each time replacing** the ball after drawing it.
- The estimate is again the proportion of black balls in your sample.

### 3. Basu's Weird Hybrid (SRSWH).

- Select one ball at a time and record its colour, mark it with an X and return it to the urn.
- If a ball drawn already has an X marked on it, then it counts as a draw, but is returned to the urn without recording its colour.
- Continue in this way until  $n$  draws have been made.
- The estimate is the proportion of black balls observed with X's.

## Comparing Sampling Mechanisms – Australian Shark Encounters

- For a population of size  $N$ 
  - there exist  $\binom{N}{n}$  samples **without replacement**
  - there exist  $\binom{N+n-1}{n}$  samples **with replacement**
  - there exist somewhere between  $\binom{N}{n}$  and  $\binom{N+n-1}{n}$  samples **with replacement but no duplicates**
- Using the Australian shark encounter population, if we take samples of size  $n = 15$ ,
  1. sampling without replacement yields a population  $\mathcal{P}_S$  of size  $M = \binom{28}{15} = 37,442,160$ .
  2. for sampling with replacement,  $\mathcal{P}_S$  is much larger, containing  $M = \binom{28+15-1}{15} = 98,672,427,616$  different possibilities.
- Using each mechanism we construct  $m = 10,000$  samples and for each sample calculate the average.

```
sharks <- read.csv("../Data/sharks.csv", header = TRUE)
```

```
popSharks <- rownames(sharks)
popSharksAustralia <- popSharks[sharks$Australia == 1]
avePop <- mean(sharks[popSharksAustralia, "Length"])
```

```
### sample size
n <- 15
### number of samples
m <- 10000

### reproducibility
set.seed(123415)
### samples without replacement
sampsWithout <- lapply(X = 1:m, FUN = function(i) {
  sample(popSharksAustralia, size = n, replace = FALSE)
```

```

})

### attribute evaluated on each sample
aveWithout <- lapply(X = sampsWithout, FUN = function(s) {
  mean(sharks[s, "Length"])
})

### samples with replacement
sampsWith <- lapply(X = 1:m, FUN = function(i) {
  sample(popSharksAustralia, size = n, replace = TRUE)
})

### attribute evaluated on each sample
aveWith <- lapply(X = sampsWith, FUN = function(s) {
  mean(sharks[s, "Length"])
})

### samples with replacement but no duplicates
aveWithUnique <- lapply(X = sampsWith, FUN = function(s) {
  mean(sharks[unique(s), "Length"])
})

### Note that in both cases, there are so many samples to choose from that
### we are not going to worry about whether we have repeated any in the m
### we have selected from M

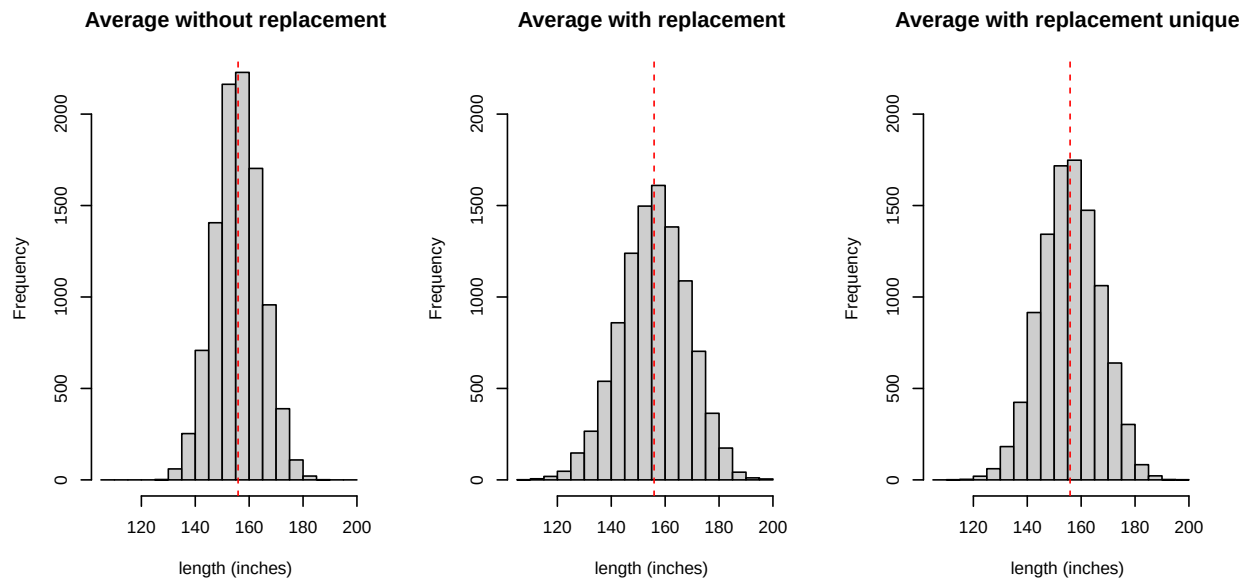
### Now prepare to plot histograms Use the same x scale in the plots
xlim <- extendrange(c(aveWith, aveWithout))
### and bins
bins <- hist(c(as.numeric(aveWithout), as.numeric(aveWith)), breaks = 30, plot = FALSE)
### And heights
ylim <- c(0, 2200)

### Without replacement
par(mfrow = c(1, 3))
hist(as.numeric(aveWithout), main = "Average without replacement", xlim = xlim,
     xlab = "length (inches)", ylim = ylim, breaks = bins$breaks, col = adjustcolor("grey",
     0.75))
abline(v = avePop, col = "red", lty = 2)

### and with
hist(as.numeric(aveWith), main = "Average with replacement", xlim = xlim, xlab = "length (inches)",
     ylim = ylim, breaks = bins$breaks, col = adjustcolor("grey", 0.75))
abline(v = avePop, col = "red", lty = 2)

### and with but no duplicates
hist(as.numeric(aveWithUnique), main = "Average with replacement unique", xlim = xlim,
     xlab = "length (inches)", ylim = ylim, breaks = bins$breaks, col = adjustcolor("grey",
     0.75))
abline(v = avePop, col = "red", lty = 2)

```



- Here is a numerical summary of these sampling distributions:

|                                       | Min   | 1st Qu. | Median | 3rd Qu. | Max   |
|---------------------------------------|-------|---------|--------|---------|-------|
| ## Without Replacement                | 129.1 | 150.3   | 155.9  | 161.9   | 186.7 |
| ## With Replacement                   | 109.9 | 147.7   | 156.3  | 164.5   | 200.0 |
| ## With Replacement but no duplicates | 111.6 | 148.5   | 156.0  | 163.7   | 196.2 |

- To compare these sampling mechanisms numerically we might also use bias, variance and MSE:

```
population = sharks[popSharksAustralia, ]$Length

# the 10,000 averages based on different sampling designs
aveWithout = unlist(aveWithout)
aveWith = unlist(aveWith)
aveWithUnique = unlist(aveWithUnique)

average = matrix(0, nrow = length(aveWith), ncol = 3)
average[, 1] <- aveWithout
average[, 2] <- aveWith
average[, 3] <- aveWithUnique

temp = rbind(apply(average, 2, mean) - mean(population), apply(average, 2, var),
             (apply(average, 2, mean) - mean(population))^2 + apply(average, 2, var))
dimnames(temp)[[1]] = c("Bias", "Var", "MSE")
dimnames(temp)[[2]] = c("Without Replacement", "With Replacement", "With Replacement but no duplicates")
round(t(temp), 4)
```

|                                       | Bias   | Var      | MSE      |
|---------------------------------------|--------|----------|----------|
| ## Without Replacement                | 0.1052 | 73.3415  | 73.3526  |
| ## With Replacement                   | 0.0969 | 153.6253 | 153.6347 |
| ## With Replacement but no duplicates | 0.1445 | 119.7212 | 119.7420 |

- What conclusions can we draw?

## Implementation of Sampling Mechanisms in R

We could implement any of the above sampling mechanisms as a single call to a factory function.

```
### This will create a sampling mechanism
createSamplingMechanism <- function(pop, method = c("withoutReplacement", "withReplacement",
  "withUnique")) {

  if (method == "withoutReplacement") {
    function(sampSize) {
      sample(pop, sampSize, replace = FALSE)
    }
  } else if (method == "withReplacement") {
    function(sampSize) {
      sample(pop, sampSize, replace = TRUE)
    }
  } else if (method == "withUnique") {
    function(sampSize) {
      unique(sample(pop, sampSize, replace = TRUE))
    }
  } else {
    stop(paste("No sampling mechanism:", method))
  }
}
```

For example, for simple random sampling without replacement on the population of all sharks, we might define a function `srswor(sampSize)` as

```
### without replacement is the default method.
srswor <- createSamplingMechanism(popSharks, method = "withoutReplacement")
```

which now allows us to generate a sample of any size containing **units selected without replacement** from the population of all sharks.

- A sample of size 5, 10, and 30

```
set.seed(341)
srswor(5)
```

```
## [1] "9" "20" "14" "63" "13"
```

```
srswor(10)
```

```
## [1] "59" "17" "29" "58" "49" "28" "7" "43" "51" "32"
```

```
srswor(30)
```

```
## [1] "27" "39" "29" "40" "25" "50" "23" "34" "11" "26" "58" "7" "17" "65" "28"  
## [16] "36" "4" "13" "53" "60" "45" "51" "42" "16" "10" "32" "63" "18" "9" "54"
```

- Similarly, for the unique units from a sample with replacement of some size:

```
set.seed(354661)  
### create the sampling mechnism  
uniquewr <- createSamplingMechanism(popSharks, method = "withUnique")  
### A sample uniquely from size 30 with replacement  
uniquewr(30)
```

```
## [1] "60" "25" "59" "10" "34" "32" "43" "26" "24" "44" "37" "41" "48" "58" "36"  
## [16] "42" "9" "29" "53" "1" "18" "12" "14" "2" "11" "28" "63" "39"
```

```
uniquewr(30)
```

```
## [1] "41" "62" "47" "50" "39" "61" "33" "51" "19" "63" "29" "65" "49" "55" "43"  
## [16] "34" "53" "38" "58" "64" "11" "31" "57" "37" "14"
```

```
uniquewr(30)
```

```
## [1] "24" "10" "61" "17" "5" "54" "58" "6" "34" "63" "59" "40" "27" "50" "29"  
## [16] "48" "44" "52" "56" "42" "18" "32" "14" "36" "19"
```

**Note** that different sample sizes can result for this method.

- The created function will only generate samples from the population `pop` which allows us to write different sampling mechanisms that might actually depend on some features of the population.

## Summary

- We introduced three different sampling mechanisms, SRSWOR, SRSWR and a hybrid of the two.
- We compared them on the basis of sampling bias, sampling variance, and sampling MSE in the context of average shark length in the Australian Shark encounter population.

## Exercises

From the Samples exercises try problems 2 and 7. For continued learning try problem 8.

### 3.2.4 Unit Inclusion Probabilities

---

#### Overview

- We introduce the concept of inclusion probabilities
- 

- The **inclusion probability** for unit  $u$  is the probability of unit  $u$  being included in the sample.

$$\pi_u = \Pr(u \in \mathcal{S}) = \sum_{\mathcal{S} \in \mathcal{P}_S} p(\mathcal{S}) \times \mathbf{I}(u \in \mathcal{S})$$

- Such inclusion probabilities are of interest to calculate in addition to  $p(\mathcal{S})$  and they may in fact be derived from  $p(\mathcal{S})$ .

- Now, consider the following random variable

$$D_u = \begin{cases} 1 & \text{if } u \in \mathcal{S} \\ 0 & \text{otherwise.} \end{cases}$$

$D_u$  takes the value 1 with probability  $\Pr(u \in \mathcal{S})$ , and it takes on the value 0 with probability  $\Pr(u \notin \mathcal{S})$ .

- The expectation of this random variable is

$$\begin{aligned} E[D_u] &= 1 \times \Pr(D_u = 1) + 0 \times \Pr(D_u = 0) \\ &= \Pr(u \in \mathcal{S}) \\ &= \pi_u \end{aligned}$$

- The variance of this random variable is

$$\begin{aligned} \text{Var}[D_u] &= E[D_u^2] - E[D_u]^2 \\ &= 1^2 \times \Pr(D_u = 1) + 0^2 \times \Pr(D_u = 0) - \pi_u^2 \\ &= \pi_u - \pi_u^2 \\ &= \pi_u(1 - \pi_u) \end{aligned}$$

- The probability that  $u$  and  $v$  are in the sample  $\mathcal{S}$  is called the **joint inclusion probability** and is given by

$$\pi_{uv} = \Pr(u \in \mathcal{S} \text{ and } v \in \mathcal{S})$$

- And we have the following relationship

$$\begin{aligned} \text{Cov}(D_u, D_v) &= E[D_u \times D_v] - E[D_u] \times E[D_v] \\ &= E[D_u \times D_v] - \pi_u \times \pi_v \\ &= \pi_{uv} - \pi_u \times \pi_v \end{aligned}$$

### Simple Random Sampling Without Replacement (SRSWOR)

- The inclusion probability is

$$\pi_u = \Pr(u \in \mathcal{S}) = \frac{1 \times \binom{N-1}{n-1}}{\binom{N}{n}} = \frac{1 \times \binom{N-1}{n-1}}{\frac{N}{n} \times \binom{N-1}{n-1}} = \frac{n}{N}$$

- The joint inclusion probability is

$$\pi_{uv} = \Pr(u \in \mathcal{S} \cap v \in \mathcal{S}) = \frac{1 \times 1 \times \binom{N-2}{n-2}}{\binom{N}{n}} = \frac{n(n-1)}{N(N-1)}$$

### Simple Random Sampling With Replacement (SRSWR)

- The inclusion probability is

$$\pi_u = 1 - \left(\frac{N-1}{N}\right)^n$$

- The joint inclusion probability is

$$\pi_{uv} = 1 - 2 \left(\frac{N-1}{N}\right)^n + \left(\frac{N-2}{N}\right)^n$$

## The Weird Hybrid Sampling Mechanism (SRSWH)

The inclusion probabilities for sampling with replacement but using only the unique units selected (i.e. the “weird hybrid” mechanism discussed earlier due to Basu) are identical to simple random sampling with replacement.

## Summary

- We introduced inclusion probabilities and derived them for for several sampling mechanisms.

## Exercises

From the Samples exercises try problem 4. For continued learning try problem 7.

## 3.2.5a Estimating Totals

---

### Overview

- Many attributes can be written as a total or as a function of a total.
  - Here we consider a general method to estimate a total for a given sampling design.
    - We derive the bias and variance of the corresponding estimator.
- 

Many attributes are either a total

$$a(\mathcal{P}) = \sum_{u \in \mathcal{P}} y_u$$

of some variate  $y_u$  observed on every unit  $u \in \mathcal{P}$ , or a function of such a total

$$a(\mathcal{P}) = f\left(\sum_{u \in \mathcal{P}} y_u\right)$$

(Recall that a variate  $y$  is any function that when applied to any unit  $u \in \mathcal{P}$  returns a value  $y(u) = y_u$ )

### Common Totals and Functions Thereof

- The **population average** is a total:

$$a(\mathcal{P}) = \sum_{u \in \mathcal{P}} \frac{y_u}{N}$$

- The **population variance** is a total:

$$a(\mathcal{P}) = \sum_{u \in \mathcal{P}} \frac{(y_u - \bar{y})^2}{N}$$

- The **population standard deviation** is a function of a total:

$$a(\mathcal{P}) = \sqrt{\sum_{u \in \mathcal{P}} \frac{(y_u - \bar{y})^2}{N}}$$

- The **number of** shark encounters from the sub-population of Australia ( $\mathcal{A} \subset \mathcal{P}$ ) is a total:

$$a(\mathcal{P}) = \sum_{u \in \mathcal{P}} I(u \in \mathcal{A})$$

- The **proportion** of units with their variate value lying in the interval  $[a, b]$  is a total:

$$a(\mathcal{P}) = \sum_{u \in \mathcal{P}} \frac{I(y_u \in [a, b])}{N}$$

- The **cumulative distribution function** (CDF) at a specific value  $y$ ,  $F_{\mathcal{P}}(y)$ , is a total:

$$F_{\mathcal{P}}(y) = \frac{1}{N} \sum_{u \in \mathcal{P}} I(y_u \leq y) = \sum_{u \in \mathcal{P}} \frac{I(y_u \leq y)}{N}$$

- A **quantile** (the inverse of the CDF) is thus an implicit function of a total:

$$Q_y(p) = \inf \{y_u : p \leq F_{\mathcal{P}}(y_u) \text{ and } u \in \mathcal{P}\}$$

- In practice, instead of  $\inf$ , we might choose to interpolate between two successive ordered values  $y_{(i)} \leq y_{(i+1)}$  whenever  $F_{\mathcal{P}}(y_{(i)}) \leq p \leq F_{\mathcal{P}}(y_{(i+1)})$ .

## The Horvitz-Thompson Estimate

- A natural (and very commonly used) estimate of a population total

$$a(\mathcal{P}) = \sum_{u \in \mathcal{P}} y_u$$

is the **Horvitz-Thompson estimate** (due to Daniel G. Horvitz and Donovan J. Thompson, 1952) defined as

$$\widehat{a(\mathcal{P})} = a_{HT}(\mathcal{S}) = \sum_{u \in \mathcal{S}} \frac{y_u}{\pi_u}$$

where the contribution for each unit in the sample is weighted inversely by  $\pi_u$ , its probability of inclusion in  $\mathcal{S}$ .

- if the probability of inclusion is small, then the weight will be high
- if the probability of inclusion is large, then the weight will be low

- Note that we use the subscript  $HT$  to distinguish the Horvitz-Thompson estimate of  $a(\mathcal{P})$  from other estimates based on the sample  $\mathcal{S}$ .

## The Horvitz-Thompson Estimator

Here we consider properties of the Horvitz-Thompson estimator (i.e., the random variable),  $\tilde{a}_{HT}(\mathcal{S})$ . Such properties inform what can be expected under repeated sampling.

### Bias, Variance, and Mean Squared Error

In what follows it will be convenient to work with the random variable

$$D_u = \begin{cases} 1 & \text{if } u \in \mathcal{S} \\ 0 & \text{otherwise.} \end{cases}$$

- Note that with this defined the Horvitz-Thompson estimator can be written as

$$\tilde{a}_{HT}(\mathcal{S}) = \sum_{u \in \mathcal{S}} \frac{y_u}{\pi_u} = \sum_{u \in \mathcal{P}} D_u \times \frac{y_u}{\pi_u}$$

The following properties of  $D_u$  that we derived earlier will also be useful:

- $E[D_u] =$
- $Var[D_u] =$
- $Cov[D_u, D_v] =$

Let us derive the **bias** of the HT estimator:

It can be shown that **variance** of the HT estimator is

$$Var[\tilde{a}_{HT}(\mathcal{S})] = \sum_{u \in \mathcal{P}} \sum_{v \in \mathcal{P}} (\pi_{uv} - \pi_u \pi_v) \frac{y_u}{\pi_u} \frac{y_v}{\pi_v} \equiv \sum_{u \in \mathcal{P}} \sum_{v \in \mathcal{P}} \Delta_{uv} \frac{y_u}{\pi_u} \frac{y_v}{\pi_v}$$

This variance can be equivalently written in the **Yates-Grundy** or the **Sen-Yates-Grundy** formulation:

$$Var [\tilde{a}_{HT}(\mathcal{S})] = -\frac{1}{2} \sum_{u \in \mathcal{P}} \sum_{v \in \mathcal{P}} \Delta_{uv} \left( \frac{y_u}{\pi_u} - \frac{y_v}{\pi_v} \right)^2$$

- The Yates-Grundy formulation provides an intuition for choosing sampling mechanisms that minimize the variance (as will be discussed further in Section 3.2.6).

Note that because the HT estimator is unbiased the **mean squared error** is simply equal to the variance.

### The HT Estimate of the Variance of the HT Estimator

- For the HT estimate we will need the following quantity

$$q_{u,v} = \Delta_{uv} \frac{y_u}{\pi_u} \frac{y_v}{\pi_v}$$

defined over  $\mathcal{P}_{uv}$  which is the population of all pairs  $(u, v)$  where  $u, v \in \mathcal{P}$ .

- The variance of the HT estimator can be written as

$$Var [\tilde{a}_{HT}(\mathcal{S})] = \sum_{u \in \mathcal{P}} \sum_{v \in \mathcal{P}} \Delta_{uv} \frac{y_u}{\pi_u} \frac{y_v}{\pi_v} = \sum_{(u,v) \in \mathcal{P}_{uv}} q_{u,v}$$

- Written this way we can see that this variance is in fact just a total!
- So...

- The HT estimate of the variance of the HT estimator is:

$$\widehat{Var} [\tilde{a}_{HT}(\mathcal{S})] = \sum_{(u,v) \in \mathcal{S}_{uv}} \frac{q_{u,v}}{\pi_{uv}} = \sum_{u \in \mathcal{S}} \sum_{v \in \mathcal{S}} \frac{\Delta_{uv}}{\pi_{uv}} \frac{y_u}{\pi_u} \frac{y_v}{\pi_v} = \sum_{u \in \mathcal{S}} \sum_{v \in \mathcal{S}} \left( \frac{\pi_{uv} - \pi_u \pi_v}{\pi_{uv}} \right) \frac{y_u}{\pi_u} \frac{y_v}{\pi_v}.$$

- Note that the sample  $\mathcal{S}_{uv}$  is obtained by sampling from the population  $\mathcal{P}_{uv}$  of all possible pairs  $(u, v)$ . The probability that any particular pair  $(u, v)$  is included in the sample is  $\pi_{uv} > 0$

- Note also that the square root of this variance estimate or HT estimate of the standard deviation is commonly referred to the **standard error** of the estimate

$$SE(\tilde{a}_{HT}(\mathcal{S})) = \sqrt{\widehat{Var}[\tilde{a}_{HT}(\mathcal{S})]}$$

- Thus, using Horvitz-Thompson estimation we are able to construct
  - an estimate of the population total *and*
  - an estimate of the variance of this estimator *and*
  - both estimators are unbiased.
- Why is this useful?

## Summary

- We introduced the Horvitz-Thompson estimate.
- We derived the bias and variance of the Horvitz-Thompson estimator.

## Exercises

- From the Samples exercises try problem 14. For continued learning try problem 16.
- Consider the simple random sampling without replacement mechanism.
  - (a) Show that the Horvitz-Thompson estimator of the population total is

$$a_{HT}(\mathcal{S}) = \frac{N}{n} \sum_{u \in \mathcal{S}} y_u$$

- (b) Show that the variance of the Horvitz-Thompson estimator in part (a) is

$$Var[\tilde{a}_{HT}(\mathcal{S})] = N^2 \left( \frac{N-n}{N-1} \right) \frac{1}{n} \left( \frac{\sum_{u \in \mathcal{P}} (y_u - \bar{y})^2}{N} \right)$$

- (c) Note that dividing  $\text{Var} [\tilde{a}_{HT}(\mathcal{S})]$  above by  $N^2$  gives the variance of the HT sample mean estimator  $\frac{1}{n} \sum_{u \in \mathcal{S}} y_u$ . The variance formula should look somewhat familiar except for a *finite population correction*  $\left(\frac{N-n}{N-1}\right)$ . Seeing this, explain the formula (divided by  $N^2$ ) in words. What if  $N \gg n$ ?

**Note:** If the last term in the R.H.S. of the equation in part (b) had the denominator  $N - 1$  instead of  $N$ , the *finite population correction* would be defined as  $\left(1 - \frac{n}{N}\right) = \left(\frac{N-n}{N}\right)$ .

## 3.2.5b Examples of Estimating Totals

### Overview

- We consider two examples of the Horvitz-Thompson Estimate

### HT Toy Example

Recall our example with a population  $\mathcal{P}$  consisting of  $N = 5$  units.

```
set.seed(341)
pop5 = round(rnorm(5), 2)
pop5 = sort(pop5)
pop5 # our population
```

```
## [1] -1.06 -0.99 -0.31  0.83  0.87
```

- Now suppose we're interested in taking a sample of size  $n = 2$ . All possible samples of size 2 are:

```
sam2 = combn(5, 2)
colnames(sam2) = paste("S", 1:10, sep = "")
sam2
```

```
##      S1 S2 S3 S4 S5 S6 S7 S8 S9 S10
## [1,]  1  1  1  1  2  2  2  3  3  4
## [2,]  2  3  4  5  3  4  5  4  5  5
```

Note that the elements of the matrix above are the indices of the samples, not the actual values of the units.

- Here we will explore Horvitz-Thompson estimation of the population mean using samples of size  $n = 2$ . The sample averages for all the samples are:

```
sam.avg <- apply(sam2, MARGIN = 2, FUN = function(s) {
  mean(pop5[s])
})
round(sam.avg, 3)
```

```
##      S1      S2      S3      S4      S5      S6      S7      S8      S9      S10
## -1.025 -0.685 -0.115 -0.095 -0.650 -0.080 -0.060  0.260  0.280  0.850
```

Recall also that we had two different sampling designs:

- d1: each sample is selected with equal probability:  $p(S) = \frac{1}{10}$ .
- d2: different samples have different probabilities of being selected. There is no intuition behind this design, we used it simply for illustration.

```
d1 = rep(1/10, 10)
d2 = 2 * (abs(apply(sam2, 2, diff)) - 1)
d2 = d2/sum(d2)
designs = rbind(d1, d2)
colnames(designs) = paste("S", 1:10, sep = "")
round(designs, 2)
```

```
##      S1  S2  S3  S4  S5  S6  S7  S8  S9 S10
## d1 0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1
## d2 0.0 0.1 0.2 0.3 0.0 0.1 0.2 0.0 0.1 0.0
```

And the sampling bias, variance and MSE associated with these designs were:

```
exp1 = sum(sam.avg * d1)
exp2 = sum(sam.avg * d2)

sam.bias = c(exp1, exp2) - mean(pop5)

sam.var = c(sum((sam.avg - exp1)^2 * d1), sum((sam.avg - exp2)^2 * d2))

designs.MSE = rbind(sam.bias, sam.var, MSE = sam.var + sam.bias^2)
colnames(designs.MSE) = c("d1", "d2")
round(designs.MSE, 5)
```

```
##              d1      d2
## sam.bias 0.00000 0.02000
## sam.var  0.26689 0.04893
## MSE      0.26689 0.04933
```

For the HT calculations below we will focus on design d2 (because HT estimation in d1 is no different than ordinary estimation via the sample mean **Why?**).

- In order to perform Horvitz-Thompson estimation we require the **marginal inclusion probabilities** for each unit in the population and the **joint inclusion probabilities** for each possible pair.
  - To calculate these we can sum  $p(\mathcal{S})$  over all samples that contain the unit (or pair of units) of interest.
  - For every sample, we determine which units are in the sample

```
inSample <- function(sam, N) {
  inSam = numeric(N) #inSam is a vector of size N
  inSam[sam] = 1
  inSam
}
pop5sam2.units = combn(5, 2, inSample, N = 5)
rownames(pop5sam2.units) = paste("unit", 1:5, sep = "")
colnames(pop5sam2.units) = paste("S", 1:10, sep = "")
pop5sam2.units
```

```
##      S1 S2 S3 S4 S5 S6 S7 S8 S9 S10
## unit1 1  1  1  1  0  0  0  0  0  0
## unit2 1  0  0  0  1  1  1  0  0  0
## unit3 0  1  0  0  1  0  0  1  1  0
## unit4 0  0  1  0  0  1  0  1  0  1
## unit5 0  0  0  1  0  0  1  0  1  1
```

Now for each unit, add up  $p(\mathcal{S})$  for each sample it appears in and that will yield,  $\pi_u$ , the **Inclusion Probabilities** for the d2 design.

```
weighted.sum <- function(x, w) {
  sum(x * w)
}
pi2 = apply(pop5sam2.units, 1, weighted.sum, w = d2)
pi2
```

```
## unit1 unit2 unit3 unit4 unit5
##  0.6  0.3  0.2  0.3  0.6
```

- To determine the **Joint Inclusion Probabilities** for d2 design we can sum  $p(\mathcal{S})$  over all samples that contain the pair of units of interest.
  - For every sample, we can determine which pairs of units are in the sample

```
inSample2 <- function(sam, N) {
  inSam = numeric(N)
  inSam[sam] = 1
  inSam = outer(inSam, inSam)
  inSam
}
# this will generate a 5x5x10 tensor
```

```
jsample.incl = combn(5, 2, FUN = inSample2, N = 5)
dimnames(jsample.incl) = list(paste("unit", 1:5, sep = ""), paste("unit", 1:5,
    sep = ""), paste("S", 1:10, sep = ""))
jsample.incl[, , 1:3]
```

```
## , , S1
##
##      unit1 unit2 unit3 unit4 unit5
## unit1     1     1     0     0     0
## unit2     1     1     0     0     0
## unit3     0     0     0     0     0
## unit4     0     0     0     0     0
## unit5     0     0     0     0     0
##
## , , S2
##
##      unit1 unit2 unit3 unit4 unit5
## unit1     1     0     1     0     0
## unit2     0     0     0     0     0
## unit3     1     0     1     0     0
## unit4     0     0     0     0     0
## unit5     0     0     0     0     0
##
## , , S3
##
##      unit1 unit2 unit3 unit4 unit5
## unit1     1     0     0     1     0
## unit2     0     0     0     0     0
## unit3     0     0     0     0     0
## unit4     1     0     0     1     0
## unit5     0     0     0     0     0
```

- Now for each pair, add up  $p(\mathcal{S})$  for each sample they appear together in and that will yield  $\pi_{uv}$ :

```
pij2 = apply(jsample.incl, c(1, 2), weighted.sum, w = d2)
pij2
```

```
##      unit1 unit2 unit3 unit4 unit5
## unit1    0.6    0.0    0.1    0.2    0.3
## unit2    0.0    0.3    0.0    0.1    0.2
## unit3    0.1    0.0    0.2    0.0    0.1
## unit4    0.2    0.1    0.0    0.3    0.0
## unit5    0.3    0.2    0.1    0.0    0.6
```

## Now let's use the Horvitz-Thompson estimate

- Recall we had samples of size  $n = 2$  and each sample was assigned a different probability  $p(\mathcal{S})$ . The sample estimate of the population average,  $a(\mathcal{P}) = \sum_{u \in \mathcal{P}} \frac{y_u}{5}$ , was defined as:

$$\text{sample average} = a(\mathcal{S}) = \sum_{u \in \mathcal{S}} \frac{y_u}{2}$$

- Recall also that the corresponding estimator was biased.

- We can use the inclusion probabilities to make our estimator unbiased:

$$a_{HT}(\mathcal{S}) = \sum_{u \in \mathcal{S}} \frac{1}{\pi_u} \times \left(\frac{y_u}{5}\right) = \frac{1}{5} \sum_{u \in \mathcal{S}} \frac{y_u}{\pi_u}$$

- Here are the HT estimates of the population mean in each of the possible samples:

```
sam.HT = apply(sam2, 2, function(s, x, wt) {
  sum(x[s]/wt[s])
}, x = pop5, wt = pi2)/5
sam.HT
```

```
##          S1          S2          S3          S4          S5          S6
## -1.01333333 -0.66333333  0.20000000 -0.06333333 -0.97000000 -0.10666667
##          S7          S8          S9          S10
## -0.37000000  0.24333333 -0.02000000  0.84333333
```

The sampling bias is then  $\sum_{\mathcal{S} \in \mathcal{P}_S} [a_{HT}(\mathcal{S}) - a(\mathcal{P})] \times p(\mathcal{S})$ :

```
sum((sam.HT - mean(pop5)) * d2)
```

```
## [1] 5.377643e-17
```

The sampling variance is then  $\sum_{\mathcal{S} \in \mathcal{P}_S} \{a_{HT}(\mathcal{S}) - E[a_{HT}(\mathcal{S})]\}^2 \times p(\mathcal{S})$ :

```
sum((sam.HT - sum(sam.HT * d2))^2 * d2)
```

```
## [1] 0.06433822
```

**Exercise:** Using the joint inclusion probabilities, calculate this sampling variance using this formula instead:

$$\sum_{u \in \mathcal{P}} \sum_{v \in \mathcal{P}} \Delta_{uv} \frac{y_u}{\pi_u} \frac{y_v}{\pi_v}$$

A comparison across sampling designs and methods of estimation is shown below.

- **Design 1 and the HT estimate**

- The HT estimate and the sample average are the same.
- Simple random sampling without replacement  $p(\mathcal{S}) = 0.1 \quad \forall \mathcal{S}$
- Estimate:

$$a_{HT}(\mathcal{S}) = \sum_{u \in \mathcal{S}} \frac{1}{\pi_u} \left( \frac{y_u}{5} \right) = \sum_{u \in \mathcal{S}} \frac{5}{2} \left( \frac{y_u}{5} \right) = \frac{1}{2} \sum_{u \in \mathcal{S}} y_u = a(\mathcal{S})$$

- **Design 2 and the sample average**

- Sampling design with  $\{p(\mathcal{S}_1), p(\mathcal{S}_2), \dots, p(\mathcal{S}_{10})\} = \{0.0, 0.1, 0.2, 0.3, 0.0, 0.1, 0.2, 0.0, 0.1, 0.0\}$
- Estimate:

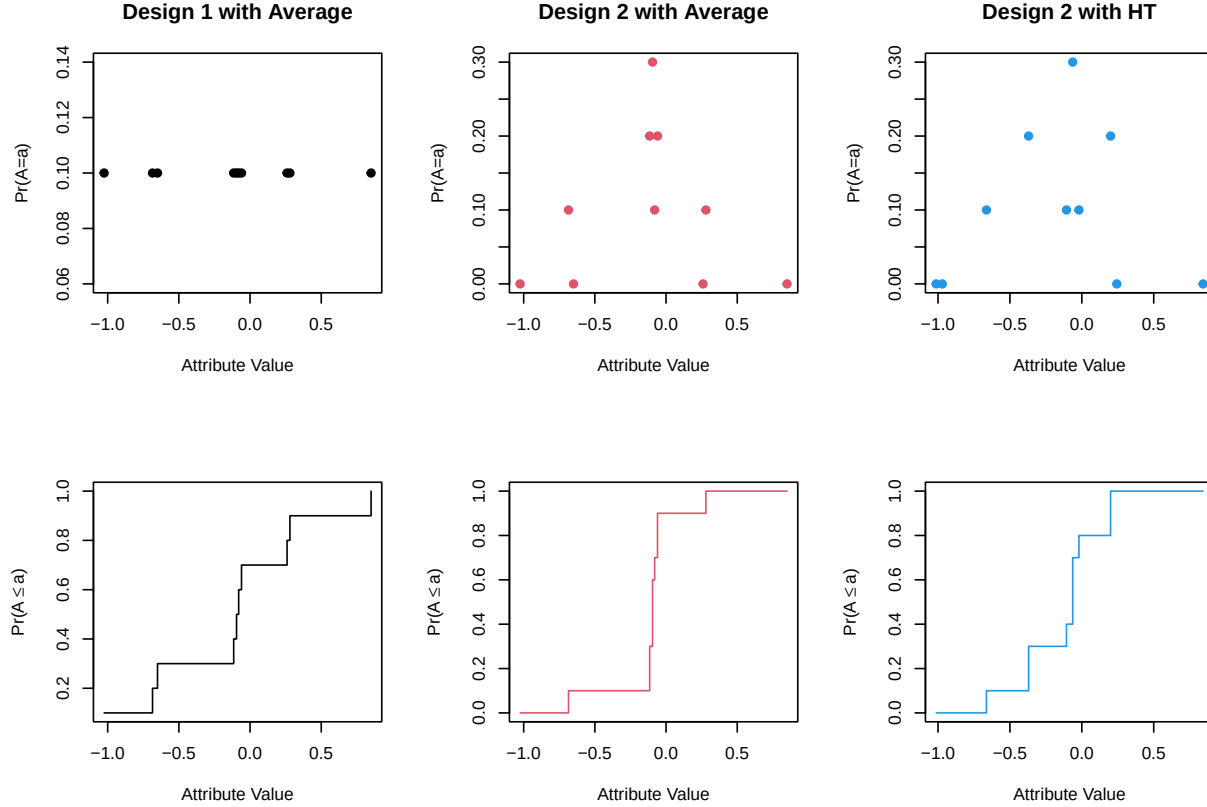
$$\text{sample average} = a(\mathcal{S}) = \sum_{u \in \mathcal{S}} \frac{y_u}{2}$$

- **Design 2 and the HT estimate**

- Sampling design with  $\{p(\mathcal{S}_1), p(\mathcal{S}_2), \dots, p(\mathcal{S}_{10})\} = \{0.0, 0.1, 0.2, 0.3, 0.0, 0.1, 0.2, 0.0, 0.1, 0.0\}$
- Estimate:

$$a_{HT}(\mathcal{S}) = \sum_{u \in \mathcal{S}} \frac{1}{\pi_u} \left( \frac{y_u}{5} \right) = \frac{1}{5} \sum_{u \in \mathcal{S}} \frac{y_u}{\pi_u}$$

- Now let's compare these three pairs of sampling design and estimation.



```
##                bias samp.var    MSE
## Design 1, Averse 0.00    0.2669 0.2669
## Design 2, Averse 0.02    0.0489 0.0493
## Design 2, HT     0.00    0.0643 0.0643
```

## HT Shark Example

Here we will explore the use of the HT estimator in the context of the Shark data. To begin with it will be useful to define the sampling mechanism as well as the marginal and joint inclusion probabilities.

- We will use sampling without replacement and hence use the `srswor` function already defined. Recall that this takes a sample of size  $n$  from the Sharks population.

```
### This will create a sampling mechanism
createSamplingMechanism <- function(pop, method = c("withoutReplacement", "withReplacement",
  "withUnique")) {

  if (method == "withoutReplacement") {
    function(sampSize) {
      sample(pop, sampSize, replace = FALSE)
    }
  } else if (method == "withReplacement") {
    function(sampSize) {
      sample(pop, sampSize, replace = TRUE)
    }
  } else if (method == "withUnique") {
    function(sampSize) {
      unique(sample(pop, sampSize, replace = TRUE))
    }
  } else {
    stop(paste("No sampling mechanism:", method))
  }
}
```

For example, for simple random sampling without replacement on the population of all sharks, we might define a function `srswor(sampSize)` as

```
sharks <- read.csv("../Data/sharks.csv", header = TRUE)
popSharks <- rownames(sharks)

### without replacement is the default method.
srswor <- createSamplingMechanism(popSharks, method = "withoutReplacement")

set.seed(341)
sample_idx <- as.numeric(srswor(10))
sharkS <- sharks[sample_idx, ]
```

- The marginal inclusion probability  $\pi_u$  when sampling without replacement is  $n/N$  for all  $u$ :

```
N <- nrow(sharks)
n <- 10
pi <- rep(n/N, N)
```

- The joint inclusion probability  $\pi_{uv}$  when sampling without replacement is  $\frac{n(n-1)}{N(N-1)}$  for any pair  $(u, v)$  and  $u \neq v$ :

```
pij <- matrix((n * (n - 1))/(N * (N - 1)), nrow = N, ncol = N)
diag(pij) <- pi
```

What follows are a variety of attributes for which the HT estimate is calculated. In each case the left number is the HT estimate of the attribute and the right number is the true population value of the attribute.

- **Total** shark length:

```
y_u <- sharkS$Length
pi_u <- pi[sample_indx]
c(sum(y_u/pi_u), sum(sharks$Length))
```

```
## [1] 9223.5 9871.0
```

- **Average** shark length:

```
y_u <- sharkS$Length/N
pi_u <- pi[sample_indx]
c(sum(y_u/pi_u), sum(sharks$Length/N))
```

```
## [1] 141.9000 151.8615
```

- **Average** victim age:

```
y_u <- sharkS$Age/N
pi_u <- pi[sample_indx]
c(sum(y_u/pi_u), sum(sharks$Age/N))
```

```
## [1] 33.7 35.6
```

- **Proportion** of shark encounters from Australia:

```
y_u <- sharkS$Australia/N
pi_u <- pi[sample_indx]
c(sum(y_u/pi_u), sum(sharks$Australia/N))
```

```
## [1] 0.4000000 0.4307692
```

- **Proportion** of shark encounters ending in a fatality:

```
y_u <- sharkS$Fatality/N
pi_u <- pi[sample_indx]
c(sum(y_u/pi_u), sum(sharks$Fatality/N))
```

```
## [1] 0.3000000 0.2615385
```

- **Proportion** of sharks with length less than or equal to 180 inches:

```
newVariate <- (sharks$Length <= 180) * 1
y_u <- newVariate[sample_indx]/N
pi_u <- pi[sample_indx]
c(sum(y_u/pi_u), sum(newVariate/N))
```

```
## [1] 0.8000000 0.7692308
```

In order to quantify the **sampling variability** of these estimators we can use the HT estimate of the variance. In what follows we calculate the estimate of the variance and the estimate of the standard deviation (also called the **standard error**) for each of the HT estimates calculated above.

But first, the following function will be helpful:

```
estVarHT <- function(sam, yu, pi, pij) {
  pi = pi[sam]
  pij = pij[sam, sam]
  delta = pij - outer(pi, pi)
  estimateVar = sum((delta/pij) * outer(yu/pi, yu/pi))
  return(estimateVar)
}
```

In each case below the left number is the estimate of the variance of the HT estimator and the right number is the estimate of the standard deviation of the HT estimator (the standard error).

- **Total** shark length:

```
y_u <- sharkS$Length
v <- estVarHT(sam = sample_indx, y_u, pi, pij)
c(v, sqrt(v))
```

```
## [1] 989834.0833    994.9041
```

- **Average** shark length:

```
y_u <- sharkS$Length/N
v <- estVarHT(sam = sample_indx, y_u, pi, pij)
c(v, sqrt(v))
```

```
## [1] 234.28026 15.30622
```

- **Average** victim age:

```
y_u <- sharkS$Age/N
v <- estVarHT(sam = sample_indx, y_u, pi, pij)
c(v, sqrt(v))
```

```
## [1] 9.665897 3.109003
```

- **Proportion** of shark encounters from Australia:

```
y_u <- sharkS$Australia/N
v <- estVarHT(sam = sample_indx, y_u, pi, pij)
c(v, sqrt(v))
```

```
## [1] 0.0225641 0.1502135
```

- **Proportion** of shark encounters ending in a fatality:

```
y_u <- sharkS$Fatality/N
v <- estVarHT(sam = sample_indx, y_u, pi, pij)
c(v, sqrt(v))
```

```
## [1] 0.01974359 0.14051188
```

- **Proportion** of sharks with length less than or equal to 180 inches:

```
y_u <- newVariate[sample_indx]/N
v <- estVarHT(sam = sample_indx, y_u, pi, pij)
c(v, sqrt(v))
```

```
## [1] 0.01504274 0.12264883
```

**Note** that calculations like the ones we've just done may also be automated using the series of general purpose functions defined in Section 3.2.5 of [this version](#) of the STAT 341 notes.

```
popSize <- function(pop) {
  nrow(as.data.frame(pop))
}
sampSize <- function(samp) {
  popSize(samp)
}

createInclusionProbFn <- function(pop, sampSize) {
  N <- popSize(pop)
  n <- sampSize
  function(u) {
    n/N
  }
}
```

```

createJointInclusionProbFn <- function(pop, sampSize) {
  N <- popSize(pop)
  n <- sampSize
  function(u, v) {
    ## Note that the answer depends on whether u and v are the same or
    ## different
    if (u == v) {
      n/N
    } else {
      (n * (n - 1)) / (N * (N - 1))
    }
  }
}

createHTestimator <- function(pi_u_fn) {
  function(samp, variateFn) {
    Reduce(`+`, Map(function(u) {
      variateFn(u) / pi_u_fn(u)
    }, samp), init = 0)
  }
}

createHTVarianceEstimator <- function(pop, pi_u_fn, pi_uv_fn) {
  function(samp, variateFn) {
    Reduce(`+`, Map(function(u) {
      pi_u <- pi_u_fn(u)
      y_u <- variateFn(u)
      Reduce(`+`, Map(function(v) {
        pi_v <- pi_u_fn(v)
        pi_uv <- pi_uv_fn(u, v)
        y_v <- variateFn(v)
        Delta_uv <- pi_uv - pi_u * pi_v

        result <- (Delta_uv * y_u * y_v)
        result <- result / (pi_uv * pi_u * pi_v)
        result
      }, samp), init = 0)
    }, samp), init = 0)
  }
}

```

Such measures of variability are most commonly used in the context of **interval estimates** (e.g., **confidence intervals**) such as:

$$\left[ a_{HT}(\mathcal{S}) - 2\widehat{SD}[\tilde{a}_{HT}(\mathcal{S})], a_{HT}(\mathcal{S}) + 2\widehat{SD}[\tilde{a}_{HT}(\mathcal{S})] \right]$$

Let's illustrate the intuition behind this formula, and let's do so in the context of the sampling distribution for the average shark length.

```

createvariateFnN <- function(popData, variate, N = 1) {
  function(u) {
    popData[u, variate]/N
  }
}
inclusionProb <- createInclusionProbFn(1:N, sampSize = n)
sharksHTestimator <- createHTestimator(inclusionProb)

N = nrow(sharks)
n = 10

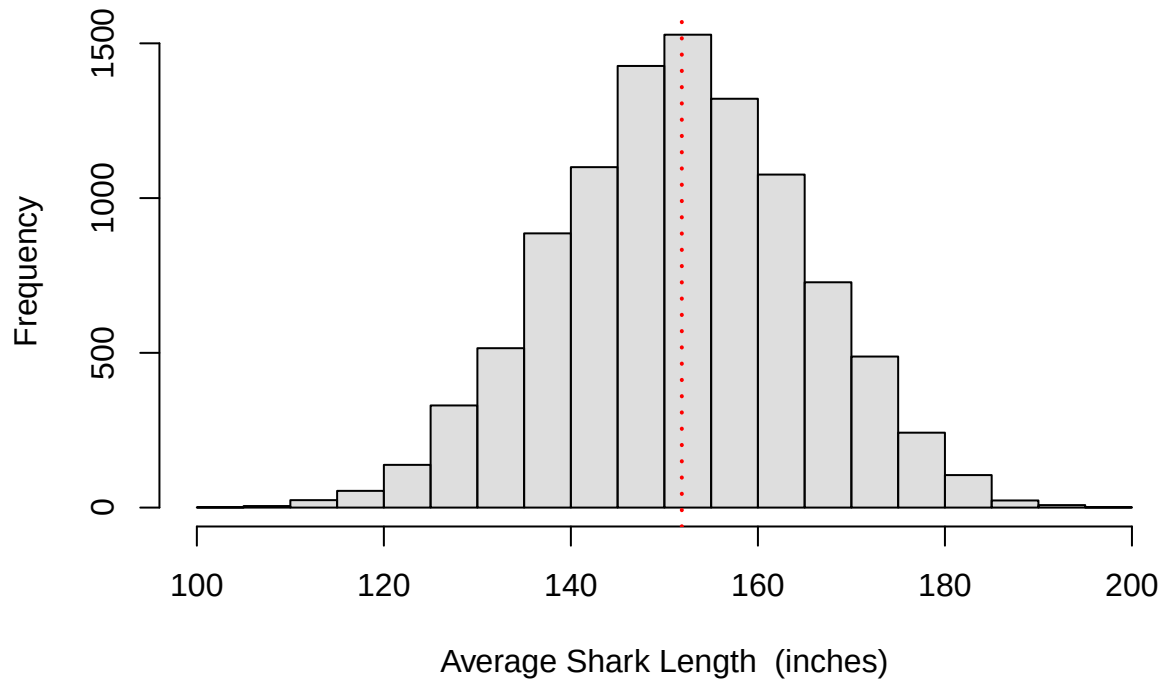
sharkAvgLength <- createvariateFnN(sharks, "Length", N = N)
popAvg <- sum(sharks$Length)/N

set.seed(341)
avgs <- mapply(function(rep) {
  sharksHTestimator(sample(N, n), sharkAvgLength)
}, 1:10000)
avgs <- as.numeric(avgs)

hist(as.numeric(avgs), col = adjustcolor("grey", alpha = 0.5), main = "Horvitz-Thompson Estimates (n = 10)",
     xlab = "Average Shark Length (inches)", breaks = 25)
### Mark the population attribute in red
abline(v = popAvg, col = "red", lty = 3, lwd = 2)

```

## Horvitz-Thompson Estimates (n = 10)



- If we use sampling without replacement and a sample size of size  $n = 10$ , how close is the sample estimate to the population value?
- To quantify this we might report some measure of dispersion for the sampling distribution, such as:
  - The interquartile range is 18.1
  - The 25<sup>th</sup> and 75<sup>th</sup> quantiles are (142.6, 160.7)
  - The 25<sup>th</sup> and 75<sup>th</sup> quantiles are the endpoints of an interval containing 50% of the HT estimates

$$\Pr(\tilde{a}(S) \in [Q(0.25), Q(0.75)]) = 0.5$$

- We might also cast a wider net and use the trimmed range where we remove the smallest and largest 2.5% (for example):
  - This trimmed range is  $Q(0.975) - Q(0.025) = 51.8$
  - The interval  $(Q(0.025), Q(0.975)) = (125.6, 177.4)$  contains 95% of the HT estimates

$$\Pr(\tilde{a}(S) \in [Q(0.025), Q(0.975)]) = 0.95$$

- **However** in order to calculate such intervals we need a reasonably good estimate of the sampling distribution, which above was obtained by drawing 10,000 samples of size  $n = 10$  from the population.
  - This is not realistic.
  - Instead we could recognize that the sampling distribution we obtained when we did this appears bell-shaped and symmetric. In other words, it could be **approximated by a normal distribution**.

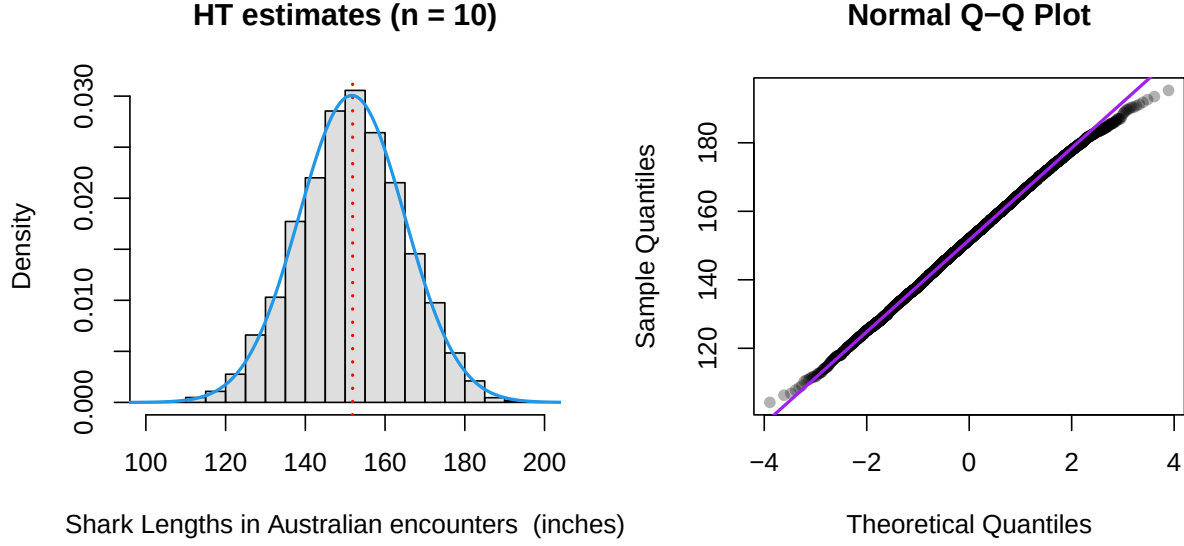
If we let  $\mu$  equal the average 151.7 and  $\sigma$  equal the standard deviation 13.3 from the population of averages as parameters in the  $N(\mu, \sigma^2)$  distribution we can approximate the sampling distribution of the HT estimator as depicted below.

```
par(mfrow = c(1, 2), oma = c(0, 0, 0, 0))
hist(avgs, col = adjustcolor("grey", alpha = 0.5), main = "HT estimates (n = 10)",
     xlab = "Shark Lengths in Australian encounters (inches)", breaks = 25,
     prob = TRUE)

### Mark the population attribute in red
abline(v = popAvg, col = "red", lty = 3, lwd = 2)

### Add the approximating normal curve
xseq = seq(10, 250, 0.01)
lines(xseq, dnorm(xseq, mean(avgs), sd(avgs)), col = 4, lwd = 2)

### Compare the observed distribution to the approximating normal
### distributions
qqnorm(avgs, pch = 16, col = adjustcolor("black", alpha = 0.3))
qqline(avgs, col = "purple", lwd = 2)
```



- As was determined using the quantiles from the samples, the interval (125.6, 177.4) contains 95% percent of the HT estimates.
- Since the histogram is approximately normal, we have that

$$0.95 = \Pr(\tilde{a}(\mathcal{S}) \in [Q(0.025), Q(0.975)]) \approx \Pr\left(\frac{\tilde{a}(\mathcal{S}) - \mu}{\sigma} \in [Z_{0.025}, Z_{0.975}]\right)$$

– Applying this result yields the following approximate interval (125.7, 177.7)

- Under an assumption of normality, an interval that contains 95% percent of the distribution's values is

$$\mu \pm Z_{1-0.05/2}\sigma$$

where  $Z_{1-0.05/2} = 1.959964 \approx 2$

- Letting  $\mu$  equal  $a(\mathcal{P})$  (the expectation of the unbiased HT estimator)

$$\mu = a(\mathcal{P}) = \mathbb{E}[\tilde{a}_{HT}(\mathcal{S})]$$

and  $\sigma$  be the standard deviation of the HT estimator

$$\sigma = \text{SD}[\tilde{a}_{HT}(\mathcal{S})] = \sqrt{\text{Var}[\tilde{a}_{HT}(\mathcal{S})]}$$

we obtain the following interval which (under an assumption of normality) approximately contains 95% of the HT estimates

$$a(\mathcal{P}) \pm 2\sqrt{\text{Var}[\tilde{a}_{HT}(\mathcal{S})]}$$

- A sample estimate of this interval is

$$a_{HT}(\mathcal{S}) \pm 2\sqrt{\widehat{\text{Var}}[\tilde{a}_{HT}(\mathcal{S})]} = a_{HT}(\mathcal{S}) \pm 2\widehat{\text{SD}}[\tilde{a}_{HT}(\mathcal{S})] = a_{HT}(\mathcal{S}) \pm 2\text{SE}[a_{HT}(\mathcal{S})]$$

This interval calculated for the average shark length based on our SRSWOR of size  $n = 10$  is

```
y_u <- sharkS$Length/N
pi_u <- pi[sample_indx]
estHT <- sum(y_u/pi_u)
sterrorHT <- sqrt(estVarHT(sam = sample_indx, y_u, pi, pij))
estHT + 2 * c(-1, 1) * sterrorHT
```

```
## [1] 111.2876 172.5124
```

**Exercise:** Calculate this interval for each of the other HT estimates investigated in this example. For each such HT estimate generate 10,000 samples and for each  $\mathcal{S}$  determine the interval

$$a_{HT}(\mathcal{S}) \pm 2\widehat{\text{SD}}[\tilde{a}_{HT}(\mathcal{S})]$$

and determine the proportion of these intervals that contain  $a(\mathcal{P})$ . Comment on your findings.

## Summary

- We reviewed the Horvitz-Thompson Estimate using two examples, and we briefly introduced the idea of a confidence interval.

## 3.2.6 Sampling Designs

---

### Overview

- We briefly discuss sampling designs.
- 

The pair  $(\mathcal{P}_S, p(\mathcal{S}))$  is called a **sampling design**

- Together they determine which samples are possible and with what probability they are selected
- The SRSWOR, SRSWR and SRSWH frameworks provide examples of different sampling designs
- The sampling design is ours to choose
  - We may choose  $\mathcal{P}_S$  so that the values  $a(\mathcal{S})$  for  $\mathcal{S} \in \mathcal{P}_S$  are constrained to be near  $a(\mathcal{P})$
  - We may choose  $p(\mathcal{S})$  so that samples  $\mathcal{S} \in \mathcal{P}_S$  that have  $a(\mathcal{S})$  close to  $a(\mathcal{P})$  have higher probability  $p(\mathcal{S})$  of being selected
  - Within the Horvitz-Thompson framework we know that

$$MSE[a_{\tilde{HT}}(\mathcal{S})] = Var[a_{\tilde{HT}}(\mathcal{S})] = -\frac{1}{2} \sum_{u \in \mathcal{P}} \sum_{v \in \mathcal{P}} \Delta_{uv} \left( \frac{y_u}{\pi_u} - \frac{y_v}{\pi_v} \right)^2$$

which provides insight into how we might best choose a sampling design.

- \* For example, if we could choose  $\pi_u \propto y_u$  then the variance (and MSE) will be zero!
- \* Perhaps there is another variate  $x_u$  that is highly positively correlated with  $y_u$  for all  $u \in \mathcal{P}$ . Then choosing  $\pi_u \propto x_u$  could reduce MSE.
- \* If we knew when  $y_u \approx y_v$  we could choose  $\pi_u \approx \pi_v$  and this might reduce MSE (e.g., stratified sampling tries to do this).

Much of survey sampling is concerned with how best to choose the sampling design  $(\mathcal{P}_S, p(\mathcal{S}))$  to reduce the MSE of an estimator (attribute) of interest.

## Summary

- We briefly discussed sampling designs.

## Exercises

From the Samples exercises try problem 15.